

8(a)(i)	photoelectric effect	B1
8(a)(ii)	electron diffraction	B1
8(b)(i)	$\lambda = h / p$	C1
	$p = 4 \times 1.66 \times 10^{-27} \times 6.2 \times 10^7$ (= $4.1 \times 10^{-19} \text{ N s}$)	C1
	$\lambda = 6.63 \times 10^{-34} / 4.1 \times 10^{-19}$ $= 1.6 \times 10^{-15} \text{ m}$	A1
8(b)(ii)	line with negative gradient throughout	B1
	curve asymptotic to both axes with non-zero λ at $v = 6.2 \times 10^7 \text{ m s}^{-1}$	B1
8(c)	(de Broglie) wavelength negligible compared with width of doorway	B1